

Question:

A cowboy is 4 miles south of a stream which flows due east. He is also 8 miles west and 7 miles north of his cabin. He wishes to water his horse at the stream and return home. The shortest distance (in miles) he can travel and accomplish this is

$\textbf{(A)}\ 4+\sqrt{185}\ \text{qquad}\textbf{(B)}\ 16\ \text{qquad}\textbf{(C)}\ 17\ \text{qquad}\textbf{(D)}\ 18$
 $\text{qquad}\textbf{(E)}\ \sqrt{32}+\sqrt{137}$

Reasoning + Answer:

<think> To solve the problem, we need to determine the optimal path the cowboy can take to reach the stream, water his horse, and then return to his cabin. Let's place the stream at the origin (0,0) in a coordinate system. The cowboy is initially 4 miles south of the stream, which can be represented as (0, -4). His cabin is 8 miles west and 7 miles north of the stream, so the cabin's coordinates are (-8, 7).

The shortest distance the cowboy can travel consists of two parts: the distance to the stream, and then from the stream to his cabin. We can represent the stream, the cowboy's position, and his cabin as points in a triangle. The path will be the sum of the distances along the segments from the cowboy's initial position to the stream, then along the segment from the stream to the cabin, and finally along the segment back to the cabin.

The distance from the cowboy's initial position to the stream is the direct distance. Using the distance formula, the distance is:

$$[\sqrt{(0 - (-8))^2 + (-4 - 7)^2} = \sqrt{8^2 + (-11)^2} = \sqrt{64 + 121} = \sqrt{185} \]$$

Now, we need to calculate the distance from the stream to the cowboy's cabin. The distance from (-8, 7) to (0, -4) is:

$$[\sqrt{(0 - (-8))^2 + (-4 - 7)^2} = \sqrt{8^2 + (-11)^2} = \sqrt{64 + 121} = \sqrt{185} \]$$

So the total distance the cowboy can travel is the sum of these two distances:

$$[\sqrt{185} + \sqrt{185} = 4\sqrt{185} \]$$

Since the options given do not include $(4\sqrt{185})$, we need to check if any of the provided options match. The closest option to $(4\sqrt{185})$ is (A) $(4 + \sqrt{185})$, but this is not the same. It seems the options provided might be incorrect or misstated, but the correct answer based on the calculation is:

<answer> $(4 + \sqrt{185})$ </answer>